

Motomachi & the Bay, Hakodate

Then-and-now aerial photographs, how the ground was made, and the disasters people chose to remember — compiled automatically from public Japanese government data.



元町・ベイエリア（函館市）

Study area	2.2 km × 1.4 km around 41.7650° N, 140.7150° E
Oldest aerial photograph	1961–1969
Aerial series in this brief	1961–1969, 1974–1978, 2024
Main natural landform	Sandbars and dunes (34%)
Former river channels	none mapped
Filled, reclaimed or drained land	10%
Former sea, river or pond	10%
Meiji-era paddies, marsh and flats	none mapped
Disaster memorials within 3 km	0

What this brief is. A reading aid for comparing the same ground across decades. It shows what the photographs and GSI's land surveys record; it does not describe current conditions, access or opening hours.

What it is not. Not a hazard assessment or a property survey. Landform classes describe typical tendencies of the ground; for present-day risk, read the official hazard map of the municipality.

Edition 2026-09-27. Aerial photographs, landform and Meiji-era lowland data: Geospatial Information Authority of Japan (GSI), processed by Japan Time Atlas. Memorial data: GSI Natural Disaster Memorials, summarised and translated by Japan Time Atlas.

The same ground, 1961–1969 and 2024



1961–1969 aerial photograph — GSI Tiles *ort_old10*, cropped and annotated by Japan Time Atlas. North is up; the frame is identical in both images.



2024 annual orthophoto — GSI Tiles *nendophoto2024*, cropped and annotated by Japan Time Atlas. Capture dates within a mosaic can differ from place to place.

How to compare. Anchor on features that rarely move — river banks, rail lines, shrine and temple grounds, main roads — then look for what changed around them. Differences in tone come from film, camera and season, not from the land itself.

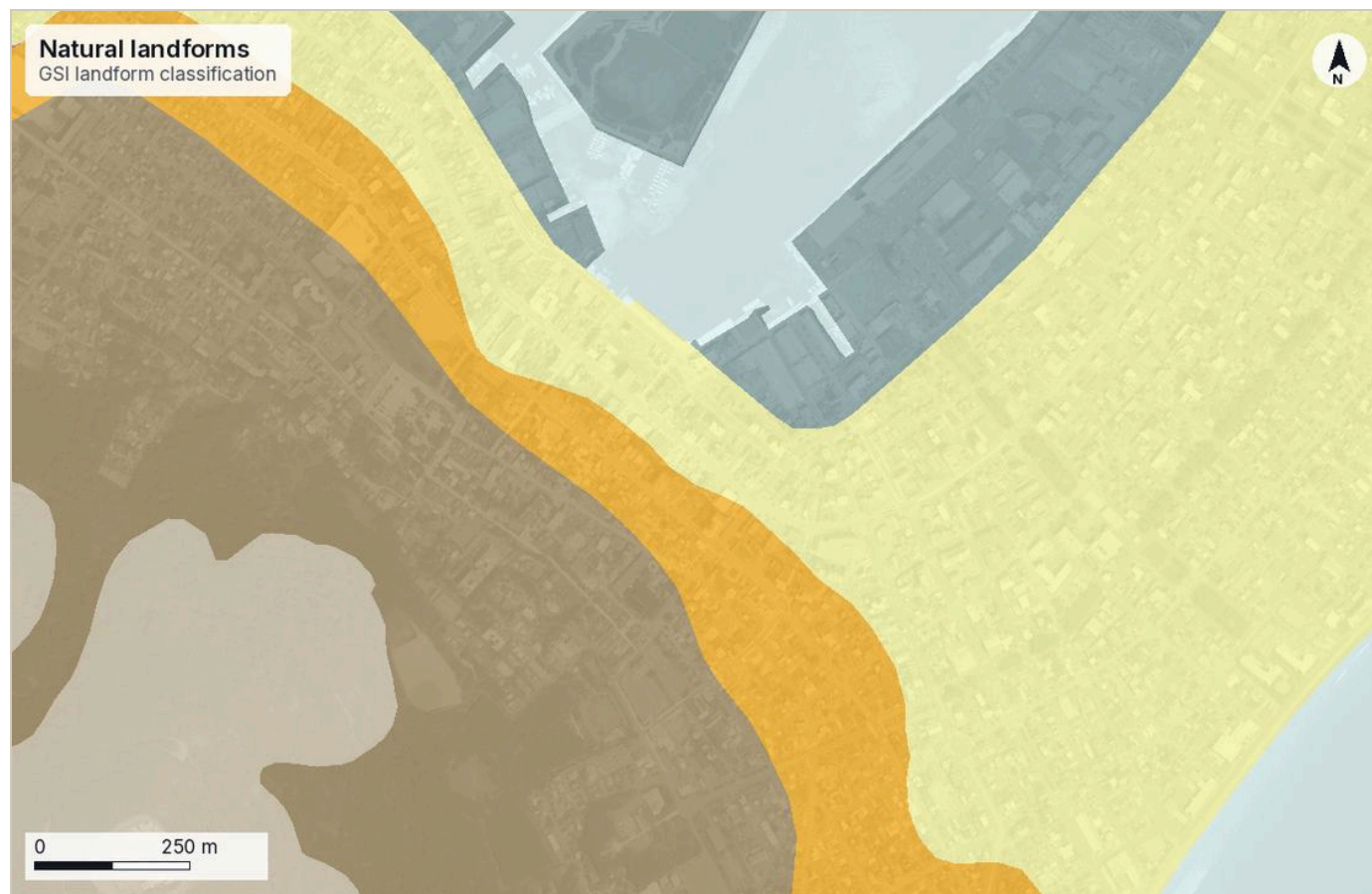
The decades in between

Every further GSI series that covers the same frame, oldest first. Each series is a compilation: photographs within one series were taken on different dates, which is why seams and changes of tone can appear inside a frame.



1974-1978 · GSI Tiles *gazo1*, processed

Rivers, shorelines and terraces



Natural landforms as classified by GSI, over a recent aerial photograph in gray. Dark blue outlines mark former river channels.

 Sandbars and dunes 34%	 Foothill deposits 26%
 Plateaus and terraces 11%	 Former water area 10%
 Water 10%	 Mountain and hill slopes 9.3%

What the main landforms mean

Sandbars and dunes (34%). Sandy or gravelly ground higher than its surroundings along present or former coasts, lakeshores and riverbanks, built by waves or wind. Usually escapes ordinary floods. Liquefaction tendency is somewhat strong on sandbars, somewhat weak on dunes and very strong at dune margins.

Foothill deposits (26%). Gentler ground below mountains and scarps, built from debris brought down by slope failures and debris flows. Debris flows are possible in heavy rain; the ground is unstable and slopes may fail in earthquakes.

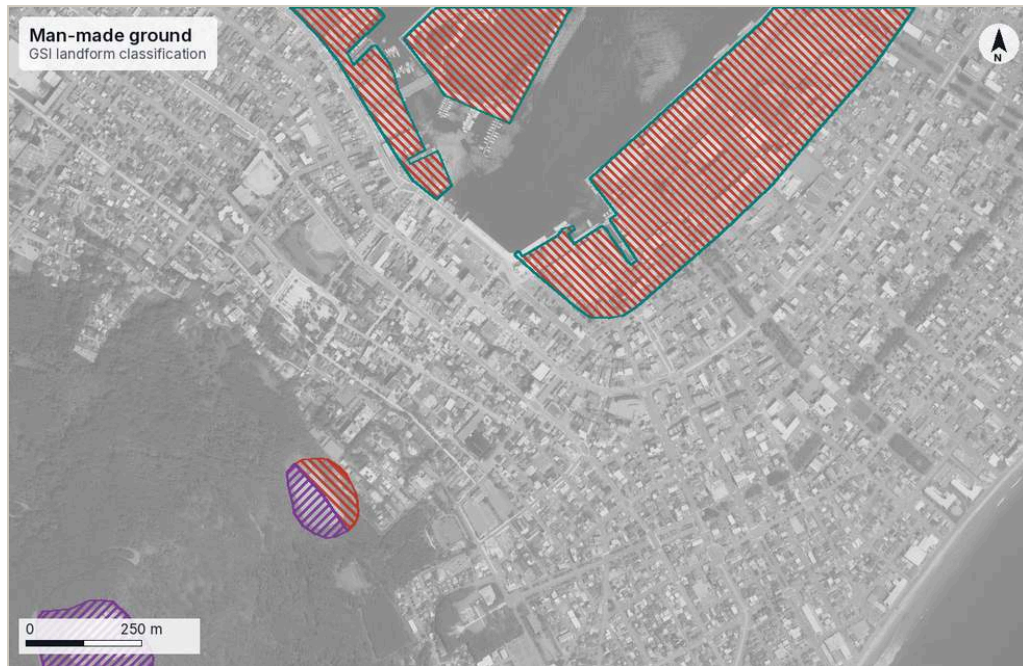
Plateaus and terraces (11%). Flat ground standing in steps above its surroundings, left behind as the land around it was eroded. River flooding is rarely a threat unless the terrace is only slightly above the river; watch the slopes at its edges. Firm ground: shaking is relatively weak and liquefaction is unlikely.

Former water area (10%). Land shown by old maps and aerial photographs to have been sea, lake, pond, reservoir or river, later turned into land. Soft ground: very strong liquefaction tendency where filled, strong where drained. Storm surge near the coast.

Mountain and hill slopes (9.3%). Ridges, valleys and fairly steep ground, including terrace-scarp slopes and volcanic ground. Heavy rain or earthquakes can trigger slope failures, debris flows and landslides.

Descriptions translated and condensed from GSI's landform-classification legend. They describe tendencies of each landform type, not the condition of any individual plot.

Fill, reclamation and the fields beneath the town



Hatching: man-made ground as classified by GSI. Teal outlines: former sea, river or pond. Dark blue outlines: former river channels.

Man-made ground

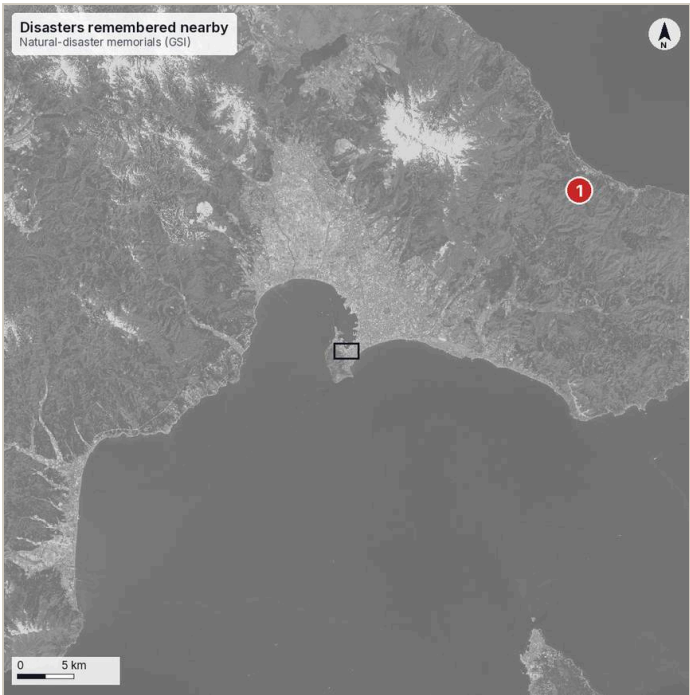
-  Filled and reclaimed land 10%
-  Cut ground 1.3%

Meiji-era lowland

GSI has not traced Meiji-era lowland for this frame.

Filled and reclaimed land (10%). Ground raised with fill, land made by dumping soil into the sea, rivers or ponds, or valleys filled in. May flood if not raised enough; on hills, rain or earthquakes can make the fill fail; in lowlands the liquefaction tendency is very strong, especially on filled sea, lake or river beds.

Memorials to natural disasters nearby



Numbered memorials; the black rectangle is the study area of this brief. Frame 60 km across.

Fewer than three memorials stand within 3 km of the study area, so the nearest ones within 30 km are listed.

Memorial	Disaster	Erected	Distance
1 Memorial to the victims of the 1973 torrential rain in Minamikayabe 遭難犠牲者慰霊碑	Torrential rain (24 Sep 1973) · flood / landslide or debris flow	1975	24.7 km

What the memorials record

1. Memorial to the victims of the 1973 torrential rain in Minamikayabe. Torrential rain fell on the southern Oshima Peninsula from the night of 23 September 1973 into the 24th. In the former town of Minamikayabe (now northern Hakodate) a record 383 mm caused floods, debris flows and slope collapses; the national highway was cut, 16 houses were destroyed, 8 people died and 11 were injured.

Source: GSI Natural Disaster Memorials (自然災害伝承碑). English names and notes are Japan Time Atlas translations and summaries of GSI's Japanese descriptions; the memorials' own inscriptions may say more. "Erected" is the year GSI records for the monument.

Where every element comes from

- **Aerial photographs** (pp. 2–3) — Geospatial Information Authority of Japan (GSI), GSI Tiles: series *ort_old10* (1961–1969), *gazo1* (1974–1978) and *nendophoto2024*. Cropped, resized and annotated by Japan Time Atlas. List of GSI Tiles: <https://maps.gsi.go.jp/development/ichiran.html>
- **Landform classification** (pp. 4–5) — GSI landform-classification vector tiles (natural and artificial landforms), compiled by GSI from its Land Condition Maps, Flood Control Terrain Classification Maps and vulnerable-landform surveys. Rendered, measured and translated into English by Japan Time Atlas. https://github.com/gsi-cyberjapan/experimental_landformclassification
- **Meiji-era lowland** (p. 5) — GSI “明治期の低湿地” tiles and legend (https://maps.gsi.go.jp/legend/lw_legend.pdf), measured and translated by Japan Time Atlas.
- **Natural disaster memorials** (p. 6) — GSI Natural Disaster Memorials (自然災害伝承碑), registered by municipalities, <https://www.gsi.go.jp/bousaichiri/denshouhi.html>; processed by Japan Time Atlas (selection by distance, English translation and summary). Memorial photographs are not reproduced.
- GSI content is used under the GSI Content Terms of Use, which follow the Public Data License v1.0 (PDL1.0, compatible with CC BY 4.0). Data retrieved 2026-09-27.
- **Outline of Japan** — Data of Japan / JapanPrefGeoJson (public domain), simplified.

Method

The study area is a fixed rectangle of 2.2 × 1.4 km. Every GSI photo series that covers at least 85% of it is included; the newest annual orthophoto covering at least 90% of it is used as “now”, with any gap filled from GSI's latest seamless mosaic. Landform shares are areas of the rectangle measured on GSI's polygons at zoom level 14, each class merged before measuring. Meiji-era shares count map pixels by legend color. Memorials are selected by straight-line distance from the center of the study area.

Limits

Historical series are mosaics of photographs taken on different dates; GSI publishes the flight date of each photograph. Landform data record conditions at the time of survey. Nothing in this brief shows present-day access, safety or opening hours.

Area Dossier. The same district with a 4–8 page English history, written from Japanese library sources and cited page by page.

Deep Research. A single hamlet or address, researched in libraries and archives.

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